MAT3032 ANSWERS TO EXERCISES 2.2

Exercises 2.2

1. (a) $C_{S_3}(\sigma)$ consists of the elements of S_3 which commute with σ, so it is $\{\iota, \sigma, \sigma^2\}$. The conjugacy class of σ is $\{\sigma, \sigma^2\}$.

   (b) $C_{Q_4}(a) = \{e, a, a^2, a^3\}$. Thus the conjugacy class of a has order $8/4 = 2$, so it consists of a and one other element. We find $bab^{-1} = a^3$, so the conjugacy class of a is $\{a, a^3\}$.

2. $\{\iota\}$ is a one-element conjugacy class.

   Conjugating σ^2 by other elements always gives σ^2, so $\{\sigma^2\}$ is a conjugacy class.

   Conjugating σ by other elements always gives σ or σ^3, so $\{\sigma, \sigma^3\}$ is a conjugacy class.

   Conjugating τ by other elements always gives τ or σ^3, so $\{\tau, \sigma^2\tau\}$ is a conjugacy class.

   If the remaining two elements were in separate one-element conjugacy classes they would be in $Z(D_4)$, which the table for D_4 shows they are not. Hence they are in the same class $\{\sigma\tau, \sigma^3\tau\}$.

   Thus the conjugacy classes in D_4 are $\{\iota\}, \{\sigma^2\}, \{\sigma, \sigma^3\}, \{\tau, \sigma^2\tau\}, \{\sigma\tau, \sigma^3\tau\}$.

   (Note: elements with the same cycle type in S_4 are conjugate in S_4 but not necessarily in D_4. However, elements which are conjugate in D_4 must have the same cycle type, e.g. σ and σ^3 are both 4-cycles.)

   $Z(D_4) = \{\iota, \sigma^2\}$ (the union of the one-element classes).

   The unions of conjugacy classes which are subgroups of D_4 are $\{\iota\}, \{\iota, \sigma^2\}$, $\{\iota, \sigma, \sigma^2, \sigma^3\}, \{\iota, \sigma^2, \tau, \sigma^2\tau\}, \{\iota, \sigma^2, \sigma\tau, \sigma^3\tau\}$ and D_4 itself. Hence these are all the normal subgroups of D_4.

3. If G is abelian then $Z(G) = G$ so $|Z(G)| = |G| = p^n \neq p^{n-1}$.

   If G is non-abelian, $|G| = p^n$ and $|Z(G)| = p^{n-1}$ then $|G/Z(G)| = p$ so $G/Z(G)$ is cyclic of order p, contradicting Proposition 2.11. Thus $|Z(G)| \neq p^{n-1}$.

4. If H is a subgroup of $Z(G)$ then it is certainly a subgroup of G. Also for all $a \in G, h \in H$ we have $ah = ha$ so $aha^{-1} = h \in H$. Hence $H \lhd G$.

5. By Proposition 2.11, $G/Z(G)$ is not cyclic. All groups of order less than 4 are cyclic so $|G/Z(G)| \geq 4$. Thus $\dfrac{|G|}{|Z(G)|} \geq 4$, so $|Z(G)| \leq \dfrac{1}{4}|G|$.

   For any $x \in G$ we know that order of the conjugacy class of x is $\dfrac{|G|}{|C_G(x)|}$.

   Now $Z(G) \subset C_G(x)$, so $|Z(G)| \leq |C_G(x)|$ and hence $\dfrac{|G|}{|C_G(x)|} \leq \dfrac{|G|}{|Z(G)|}$.

   Thus if $\dfrac{|G|}{|Z(G)|} = n$ then $\dfrac{|G|}{|C_G(x)|} \leq n$.

6. We have $b = gag^{-1}$ for some $g \in G$. If $x \in \text{Fix}(a)$ then $a(x) = x$, so $g^{-1}bg(x) = x$, so $b(g(x)) = g(x)$ and thus $g(x) \in \text{Fix}(b)$. Hence g maps $\text{Fix}(a)$ into $\text{Fix}(b)$.

   As the action of g permutes the set X, it is a bijection so $|\text{Fix}(a)| = |\text{Fix}(b)|$.

7. (a) As $H \cap J = \{e\}$, each element of HJ has a *unique* expression in the form hj, so the given rule is well-defined.
   $ghg^{-1} \in H$ since $H \lhd G$, so $ghg^{-1}j \in HJ$. $\quad e(hj) = ehe^{-1}j = hj$.
   $g_1(g_2(hj)) = g_1(g_2hg_2{}^{-1}j) = g_1g_2hg_2{}^{-1}g_1^{-1}j = (g_1g_2)h(g_1g_2)^{-1}j = g_1g_2(hj)$, so we have a group action of G on HJ.

   (b) $G_{hj} = \{g \in g \ : \ ghg^{-1}j = hj\} = \{g \in g \ : \ ghg^{-1} = h\} = C_G(h)$.
   $\widetilde{G}_{HJ} = \{g \in G \ : \ ghg^{-1}j = hj \text{ for all } hj \in HJ\}$
   $= \{g \in G \ : \ gh = hg \text{ for all } h \in H\}$, the set of all elements of G which commute with *every* element of H. (This is not the same as the normaliser of H in G.)

8. $253 = 11 \times 23$. By the same reasoning as in Examples 2.2.1, $|Z(G)| = 1$ so their is only one one-element conjugacy class, and the other classes have order 11 or 23. If there are m and n of these respectively then $11m + 23n = 252$. The only integer solution is $(m, n) = (2, 10)$ so there is one class of order 1, two of order 11 and ten of order 23, giving 13 classes altogether.

   A normal subgroup must be a union of conjugacy classes, including the class $\{e\}$ of order 1, and its order must divide 253. The only way of achieving this is $11 + 11 + 1 = 23$, i.e. any non-trivial proper normal subgroup has order 23 and is the union of two 11-element classes and the 1-element class.

9. Let $X = \{\text{subgroups of } G\}$. Then $gHg^{-1} \in X$ for all $H \in X$.
   $e(H) = eHe^{-1} = H$ and $g_1(g_2(H)) = g_1(g_2(H))g_1^{-1} = g_1g_2Hg_2^{-1}g_1^{-1}$
   $= (g_1g_2)H(g_1g_2)^{-1} = (g_1g_2)(H)$.

   Thus we have an action of G on the set X of all its subgroups.

   $\text{Fix}(G) = \{H \in X : \ gHg^{-1} = H \text{ for all } g \in G\} = \{\text{normal subgroups of } G\}$.

10. (a) $hH = Hh = H$ for all $h \in H$, so $H \subset N_G(H)$. As H is a subgroup of G, it is a subgroup of $N_G(H)$.
    By definition of $N_G(H)$, every left coset of H in $N_G(H)$ equals the corresponding right coset, so $H \lhd N_G(H)$.

    (b) $H \lhd G \Leftrightarrow gH = Hg$ for all $g \in G \Leftrightarrow N_G(H) = G$.

    (c) The subgroups which are conjugate to H (including H itself) form the orbit of H under this action. The orbit-stabiliser equation gives $|G| = |\mathcal{O}(H)|.|N_G(H)|$, from which the result follows.

    (d) The number of elements in the union of the subgroups conjugate to H is less than the sum of their orders, since any two intersect in at least the identity. There are $|\mathcal{O}(H)|$ subgroups conjugate to H, so the sum of their orders is $|\mathcal{O}(H)|.|H|$, which equals $\dfrac{|G|}{|N_G(H)|}.|H|$ by (c). As $H \subset N_G(H)$ we have $|H| \leq |N_G(H)|$, so the union of the conjugates of H has order less than $|G|$ and hence their union cannot equal G.